Abstract:

The translation from competing in college basketball to the National Basketball Association (NBA) has long been captivating for NBA scouts, analysts, and franchises. Our project will investigate the accuracy of how college basketball statistics can accurately predict career performance in the NBA, using individually chosen metrics such as win shares per 48 minutes (WS/48) and the player efficiency rating (PER). Using datasets from “Basketball-Reference”, we will use regression-based and machine learning methods to pinpoint which college-level metrics have the most carryover and predictive significance for long-run performance in the NBA.

Beyond just our core question, we are aiming to explore a few secondary angles. First, we will test whether success (and the predictability of it) is different among different positions. Second, we will assess if the round in which a player is drafted has strong predictive signalling. Third, we will explore whether the age at which players are drafted heavily influences the college-to-NBA transition. This is an important relationship to explore because younger players are drafted on projected development rather than proven college production. In this question, we distinguish early entrants such as freshmen and sophomores from upperclassmen.

The results we obtain will hopefully indicate that college statistics carry an average, but meaningful signal significance for NBA performance, with efficiency metrics outperforming the volume statistics. Predicting accuracy will likely vary by position, draft round will likely add additional signals, and the age at entry significantly affects the likelihood of early success in the NBA. These findings will have practical implications for future draft evaluation and player development frameworks.

Introduction:

Every year in June, NBA teams make potentially franchise-changing decisions at the NBA draft based solely on projection information and opinions. A player's college career provides the best record of their skill and talent at the time of the draft. Nonetheless, effectively being able to accurately predict the translation from college production into the league has been historically difficult for NBA franchises. College heavily differs from the NBA in terms of game speed, intensity, and competition level. This could raise the question about how much signal significance college statistics actually have. Despite this, college scouting and player analytics have grown substantially, and the question of whether models can improve on traditional scouting judgment has become practically important.

Our project addresses our main question: Can NBA career performance be accurately predicted given a set of college player stats? Our primary outcome variables of interest are win shares per 48 minutes (WS/48) and the player efficiency rating (PER). WS/48 evaluates player contribution to his team's wins on a per-possession basis, while PER totals per-minute production into a single efficiency index. Together, they give a solid perspective on player impact, making them a robust dual criterion for evaluating college prospects.

We will define and explore several unique scoring statistics along with how they translate to the league. Scoring metrics such as true shooting percentage and points per possession translate more reliably than raw scoring volume, since offensive roles can largely increase or decrease. Next, rebounding rates, capture athleticism and effort tendencies that usually persist across both levels. Playmaking indicators reflect decision-making quality and vision on the court. Defensive indicators may signal athleticism and positional instincts. Of all the metrics so far, defense is likely to have the most statistical “noice”. Finally, usage rate provides important context, distinguishing high-volume producers from efficient role players who accumulate strong efficiency numbers in limited opportunities.

Beyond the primary task, this project will pursue three secondary questions. The first question explores whether the college-to-professional transition is equally predictable for guards, forwards, and centers. Role demands obviously differ across positions, and the extent to which college statistics generalize them may vary accordingly. For example, a guard who scores more efficiently in college will face different challenges than a center who dominates rebounding in a smaller college in a weaker conference.

The second question examines the impact of the specific round a player was drafted in. Draft position is a form of aggregated judgment, which incorporates scouting reports, player measurements, college stats, and other intangibles. It is therefore possible that the draft round contains information that is directly related to collegiate success. Testing whether draft round adds predictive signal beyond college statistics explicitly addresses how much additional information scouts contribute to a purely statistical baseline.

The third question investigates how the age of a draftee can impact their NBA success. Collegiate freshmen and sophomores who enter the draft early are typically valued at their ceilings rather than their college production. Their college stats carry different weight than those of upperclassmen who have peaked within college. A framework is used to test whether the college-to-NBA transition is attenuated for younger draftees and whether accounting for age at entry improves the overall fit of our model.

Following the bulk of this project, there are supplemental sections that further explore the questions we ask and our model itself. Section 3 reviews our methodology for our analytics, draft outcome prediction, and the college-to-professional translation problem. Section 4 explains the data sources, sample construction criteria, and all key variables. Section 5 details the statistical methods employed, including OLS regression, regularized regression approaches for variable selection, and the results generated from our model and analysis. Section 6 concludes with a summary of key takeaways and practical implications for NBA draft evaluation.